



## Original article

## Leveraging part-and-sensitive attention network and transformer for learner engagement detection

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## ABSTRACT

The assessment of student engagement levels in the context of online learning represents a pivotal challenge precipitated by the rapid and substantial progression of digitalization within the realm of education. This paper introduces a hybrid architecture, termed DTransformer, for engagement detection. Specifically, the architecture consists of two branches: a PANet and a STformer architecture. Extensive experiments are conducted on DAiSEE and EmotiW-EP dataset to obtain competitive performances for student engagement detection. The result shows that on DAiSEE dataset, the accuracy is 64% on the test set, whereas on the EmotiW-EP dataset, the MSE is 0.0729. The performances of our method has demonstrates the effectiveness of the proposed method.

## 1. Introduction

Online learning is not limited by time and geography, and can be done synchronously or asynchronously on different devices. Therefore, it is widely used in modern education situation as a complementary approach to traditional classroom education. Online education, however, brings issue in the process of teaching that learners may not receive guaranteed learning outcomes as planned or expected by educators partially due to a lack of effective supervision of their learning behaviors as well as real-time interaction between instructors and learners during the knowledge acquisition process. It is, therefore, of necessity to monitor learners' attention status in a comparatively effective way so as to ensure the learning quality of online learners. Research data on attention detection embraces learners' images [1], videos [2], audios [3], and electrocardiograms (ECG) [4]. Compared with biological sensors, network cameras are lower in costs and more convenient for collecting learners' in-class performance data, and thus employed in most recent research works on learners' attention detection data collection. Computer vision technology is utilized afterwards for engagement detection [4,5].

Despite the performances obtained, existing methodologies still reveal several limitations. Conventional approaches pertaining to computer vision demonstrate commendable performances under a subset of standard scenarios; however, their efficacy diminishes considerably when confronted with complex situations characterized by obstructions or fluctuations in posture. In such instances, these methods are compelled to undertake an elaborate preliminary processing phase to accurately register facial features, which is inherently time-intensive, or

alternately, they may fail to function directly. Contemporary techniques grounded in deep learning architecture have made strides in enhancing algorithmic robustness, culminating in remarkable achievements in the domain of engagement prediction. Yet, the process of feature extraction in these techniques necessitates additional procedural steps and remains contingent upon the preexisting knowledge of human operators, thereby reducing operational efficiency. Additionally, the prevailing scarcity of publicly accessible data sets severely constrains the scope of experimental evaluation to singular data sets for the majority of studies, thereby curbing the capacity to substantiate their algorithmic robustness comprehensively.

In general, computer vision-based methods for engagement detection of students can be divided into image-based and video-based engagement detection methods. The former detects engagement based on individual frames extracted from a single image or video, with the main limitation being the utilization of only spatial information from a single frame while engagement detection involves spatio-temporal emotional behaviors. Therefore, this study focuses on video-based learner engagement detection. Video-based engagement recognition can be divided into end-to-end detection methods and feature-based detection methods. For the end-to-end engagement detection method, consecutive raw video frames are fed into convolutional Neural Networks (CNNs), and then recurrent neural networks are used to regress the attention level [6]. In the feature-based method, manual features are extracted from video frames, and then analyzed using recurrent neural networks or machine learning methods to output the attention

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level [7]. Existing attention recognition research uses various features, including such high-level features as gaze direction, head pose, and facial action units, as well as low-level features as LBP-TOP [8] and Gabor features [9]. It is found that the feature extraction in attention recognition relies heavily on facial features. Currently, there are two main approaches for facial feature extraction: one is to use Convolutional Neural Networks (CNNs) to extract facial spatial features. But this approach cannot directly extract fine-grained facial features; the other approach is based on the extraction of a single manual feature or a combination of multiple features. But most research of this kind tend to focus on merely a few features without a fully consideration of features like facial appearance, head pose, gaze angle, blink rate, etc. In attention recognition, how to effectively combine the coarse-grained facial feature information extracted by deep networks with fine-grained behavioral features including head pose still remains a problem that has not been deeply explored yet.

To address the above limitations, a hybrid architecture, termed DTransformer, is proposed. DTransformer consists of two branches. The first branch is a part attention network (PANet), which acquires attention weights for individual partial features through an end-to-end learning process, subsequently amalgamating all features into a representation featuring global content reference. This methodology aims to accurately emphasize features that play a significant role in the prediction of depression. The second branch is a spatial-temporal transformer (STformer), which can learn the sequential representations with multi-head self-attention for engagement detection. Systematic and extensive experimental validations are undertaken to demonstrate that the proposed architecture obtains excellent performance for engagement detection.

Overall, the main contributions can be concluded below:

1. We propose a hybrid architecture, i.e. DTransformer, which not only models the 2D features by PANet, but also learns the 3D features via STformer for engagement detection.
2. In STformer, the spatial-temporal features are learned. The advantage of this branch is to learn the spatial-temporal information via a 3D transformer for engagement detection.
3. Extensive experiments are conducted on DAiSEE dataset to obtain competitive performances for engagement detection.

## 2. Related works

Since the 1980s, the topic of learner engagement detection has been extensively studied in education field. Initially, the objective of research as such was to enhance learners' involvement in educational activities. Traditionally, three categories of methods have been employed to measure learners' engagement: (1) self-reports, (2) observational checklists and rating scales, and (3) automated measurements [9]. The two methods [10,11] are not suitable for online learning due to their heavy reliance on subjective evaluations by learners or observers, lacking temporal resolution. Consequently, researchers have attempted to design automated measurement techniques for learner engagement detection.

Coccea et al. [12] and Banet et al. [13] examine learners' engagement by analyzing their interactions within online classes using an E-learning system. However, these approaches heavily rely on fixed evaluation indicators and are prone to significant biases when applied in real-world scenarios. An alternative method for automated engagement detection is based on physiological signals, such as electroencephalography (EEG), blood pressure, heart rate, or galvanic skin response. These signals are measured using specialized sensors [14–16]. Nonetheless, the high cost and potential disruption caused by these sensors pose practical challenges for their implementation in educational environment.

Automated engagement detection has primarily relied on computer vision techniques, owing to its convenient data capturing and non-intrusive assessment process. Face-based methods, in particular, have

garnered significant attention due to their ability to directly capture human emotions and their intuitive connection with engagement. For instance, Grafsgaard et al. [17] utilize the computer expression recognition toolbox (CERT) [18] to detect frequently occurring action units (AUs) and infer learners' status based on the activation of these AUs. Similarly, Bosch et al. [19] employ AUs along with face orientation and position as features to categorize learners' states into five distinct categories. Saneiro et al. [20] took a more comprehensive approach by combining AUs, 2D facial points, 3D head poses, and body movements to analyze learners' engagement. These AU-related methods offer the ability to reflect specific emotional states, which can then be mapped to engagement levels using certain predefined rules. However, these methods exhibit limited robustness in real-world scenarios and fail to accurately predict detailed levels of engagement.

Alternatively, researchers have explored the extraction of appearance features to train classifiers for engagement prediction. Whitehill et al. [9] evaluate engagement across four levels using multiple facial features and classifiers. Kamath et al. [21] utilize the histogram of oriented gradients (HOG) [22] and propose an instance-weighted multiple kernel learning support vector machines (SVM) method to enhance prediction performances. Monkaresi et al. [14] fuse multiple features, including AUs, LBP-TOP [8], and heart rate, to train learner-independent engagement detectors to improve accuracy, albeit utilizing small data sets. However, hand-crafted facial features show potential for better generalization performance, it remains challenging to design universally applicable patterns suitable for diverse conditions.

Deep learning techniques have been proven to yield excellent performances in computer vision tasks. However, various versions of deep learning techniques are adopted. For example, Nezami et al. [1] trained a simple Convolutional Neural Network (CNN) to recognize engagement in private image sets. Schulc et al. [23] propose a CNN-LSTM approach to detect engagement and non-engagement states in commercials. However, most engagement evaluation datasets available are either limited in size or lack public access, thus hindering the comparison of different algorithms and constraining progress in this field. To address these issues, Gupta et al. [6] introduce the Dataset for affective states in E-environments (DAiSEE), offering five baselines for engagement recognition. Zhang et al. [24] optimize the Inflated 3D Convolutional Network (I3D) and achieve the highest accuracy in binary engagement classification. Huang et al. [25] propose the Deep Engagement Recognition Network (DERN) to determine the degree of engagement, achieving state-of-the-art performance on DAiSEE. Additionally, learner engagement recognition has emerged as a significant challenge in emotion recognition in the Wild (EmotiW) [26,27] since 2018, fostering advancements have been seen in the field. Niu et al. [28] devise the gaze-aU-pose (GAP) feature and employ a recurrent neural network with a gated recurrent unit (GRU) layer [29] as an engagement predictor. Yang et al. [30] develop a multimodal regression model based on a multi-instance mechanism and LSTM, attaining the best results in the EmotiW 2018 sub-challenge. In [31], the authors investigate two well-established methodologies, namely Joint Adaptation Network (JAN) and adversarial domain adaptation utilizing Wasserstein distance, to address this particular task. The task involved formulating a binary classification problem, where various base models such as ResNet and I3D were employed. The most favorable outcome was achieved by employing the JAN network, displaying an accuracy of 68% and an F1-score of 0.80 when using the RGB-I3D network as the underlying model. Conversely, the adversarial domain adaptation approach achieved an accuracy of 71% and an F1-score of 0.82, with ResNet-50 serving as the feature extractor for predicting learner engagement within the classroom. In [32], the study introduces a novel model of spatio-temporal architectures comprising: a composite EfficientNet B7 and Temporal Convolution Network (TCN), an amalgamated EfficientNet B7 with Long Short-Term Memory (LSTM), and an integrated EfficientNet B7 in conjunction with Bidirectional LSTM. The role of EfficientNet B7 within these constructs was to facilitate

the extraction of salient features, whereas the temporal characteristics of the data sequences were discerned using TCN, LSTM, and Bidirectional LSTM. Upon evaluation using the DAiSEE dataset, these novel models demonstrated enhanced accuracy metrics, outstripping the performance of extant cutting-edge models for categorizing four levels of learner engagement. In [33], the study pertained a model to delineate the engagement levels of individuals as captured by an RGB camera, through analyzing parameters such as the individual's head orientation, ocular tracking, facial dynamics, and proxemics concerning the camera. Utilizing a methodology steeped in the extraction of visual characteristics coupled with the employment of a Random Forest algorithm, renowned for its efficacy in machine learning, facilitated the crafting of the engagement detection apparatus. The said apparatus was meticulously programmed to discern and categorize engagement into quadruplicate tiers. The efficacy of the apparatus was rigorously evaluated using the DAiSEE dataset, exhibiting auspicious outcomes by an F1-score metric of 0.861. However, these studies focusing on engagement prediction in online learning encounter two challenges. Despite excellent performance on individual datasets, these methods are yet to be evaluated on multiple datasets, making it arduous to establish their robustness. Furthermore, the current scale and quantity of datasets available for engagement prediction are still inadequate compared to other video datasets [34–36].

Overall, learner engagement is a multi-faceted concept encompassing the active commitment of learners to participate in their educational journey. Based on the technology, learners focus on the learning material and classroom activities. They actively listen to teachers, engage in discussions, and show interest in the subject matter. Meanwhile, engaged learners consistently complete their homework, projects, and other assigned tasks. They understand the importance of these activities in reinforcing their learning and developing skills. In addition, learners recognize the relevance and benefits of their education. They see a connection between what they are learning and their personal goals, future careers, and overall life improvement. This intrinsic motivation drives their commitment to excel academically.

### 3. Our method

#### 3.1. Architecture overview

Fig. 1 overviews the proposed architecture. The architecture consists of two branches: the first is a DCNN-architecture to extract the spatial features. For the first branch, the patch-based attentional features are extracted. In addition, a transformer is adopted to extract the sequential features. Finally, a fully connected layers is used to fuse the discriminative features to perform the final engagement prediction.

#### 3.2. PANet

In our work, ResNet-18 is adopted to extract the patch-based features. Concretely, the feature map is segmented into 5 local patches. It is observed that each of these localized maps will capture more complex feature representations via the process of convolution. However, a discordance arises, as the dimensions of these localized feature maps, at  $H/2 \times W/2$ , are in-conformed with the  $H \times W$  dimensional prerequisites incumbent upon the fifth Convolutional Layer (Conv5). This necessitates the implementation of an upsampling procedure, wherein the dimensions of the feature maps are augmented from  $7 \times 7$  to  $14 \times 14$ , in order to achieve dimensional parity. To perform the extraction of localized feature information, it is pivotal to integrate the overarching feature distribution by incorporating the global feature map. The cumulative outcome manifests in a set of  $n + 1$  partitioned feature maps; a concatenation of the original and the  $n$  elevated local feature maps. These maps collectively instantiate  $n + 1$  parallel branches within the network architecture, designated to facilitate the detection

of engagement features. Within this proposed framework, each individual branch traverses the Conv5 convolutional stream to further refine and extract high-level feature representations. An efficiency gain is actualized through the shared parameters of this convolutional operation, efficaciously diminishing the model's computational complexity. Such a strategy also serves to mitigate potential overfitting risks associated with engagement classification tasks.

For the purpose of clear understanding, let  $\mathbf{P}$  represent the composite set of partitioned feature maps, with  $\mathbf{P}_0, \mathbf{P}_1, \dots, \mathbf{P}_n$  represent the original feature map, and local feature maps, respectively.  $\gamma$  represents the Conv5 convolutional operation. Then the high level features  $F_{high}$  can be expressed as:

$$F_{high} = [\mathbf{FF}_0, \mathbf{FF}_1, \dots, \mathbf{FF}_n] \\ = [\gamma(\mathbf{P}_0, \theta), \gamma(\mathbf{P}_1, \theta), \dots, \gamma(\mathbf{P}_n, \theta)] \quad (1)$$

Then the feature map undergoes a division into  $n + 1$  branched networks, serving to extract both local and global features. Part attention mechanism is employed across each branch, directing the model's focus towards the most significant aspects of the part features. Part attention, functioning as a self-attention mechanism, takes the part feature  $\mathbf{FF}$  as its input and engages the fully connected (FC) layer and the sigmoid function to compute the weights associated with part features. The weight of the  $i$ th part feature is determined using the equation:

$$\beta_i = \text{Sigmoid}(\mathbf{FF}_i^T \mathbf{Pa}) \quad (2)$$

where  $\beta_i$  is the weight assigned to the  $i$ th input,  $\mathbf{Pa}$  denotes the FC parameters,  $\mathbf{FF}_i$  represents the  $i$ th part feature.

To condense the representation effectively, we concatenate the computed weights with the corresponding part features. This process is mathematically expressed as:

$$\mathbf{F}_{part} = \frac{1}{\sum_{i=0}^n \beta_i} \sum_{i=0}^n \beta_i \mathbf{FF}_i \quad (3)$$

where  $\mathbf{F}_{part}$  represents the combination of all part features, serving as input to the subsequent stage's relation attention module. Importantly, it encapsulates the content derived from all parts, thereby encompassing a more comprehensive and information-rich representation. The input provided to the part attention mechanism comprises  $n + 1$  high-level part features, each sized at  $7 \times 7 \times 512$ . By employing the attention module, a weighted feature of size  $7 \times 7 \times 512$  is derived, reflecting the outcome of the attention-driven process.

#### 3.3. Spatial-temporal transformer

In this section, we briefly describe the spatial-temporal transformer for engagement detection. As illustrated in Fig. 1, spatial-temporal transformer includes a 3D-CNN architecture, a patch embedding tokenizer,  $N \times$  transformer. Drawing inspiration from the investigation conducted in Ref. [37], we implement a shallow 3D-CNN architecture aimed at capturing preliminary coarse local spatio-temporal characteristics. This design choice contributes to expedited convergence rates and facilitates a more distinct subsequent global self-attention mechanism. Specifically, the stem architecture consists of three convolutional blocks utilizing kernel sizes of  $1 \times 5 \times 5$ ,  $3 \times 3 \times 3$ , and  $3 \times 3 \times 3$ , respectively. Each convolutional operation is accompanied by sequential batch normalization (BN), rectified linear unit (ReLU), and maxpool functions. Notably, the pooling layer reduces the spatial dimensions by half. Consequently, for an input RGB facial video  $\mathbf{X} \in \mathbb{R}^{3 \times T \times H \times W}$ , the output of the 3D-CNN  $\mathbf{X}_{3DCNN} = \mathbf{F}_{3DCNN}(\mathbf{X})$ , where  $\mathbf{X}_{3DCNN} \in \mathbb{R}^{D \times T \times H/8 \times W/8}$ , and  $D, T, W$ , and  $H$  represent channel, sequence length, width, and height dimensions, respectively. Subsequently,  $\mathbf{X}_{3DCNN}$  undergoes partitioning into spatio-temporal tube tokens, denoted as  $\mathbf{X}_{tube} \in \mathbb{R}^{D \times T \times H' \times W'}$ , achieved through the tube tokenizer  $\mathbf{F}_{tube}(\mathbf{X}_{3DCNN})$ .

In our work, the coarse feature  $\mathbf{X}_{3DCNN}$  undergoes partitioning into non-overlapping tube tokens through the  $\mathbf{F}_{tube}(\mathbf{X}_{3DCNN})$  procedure.

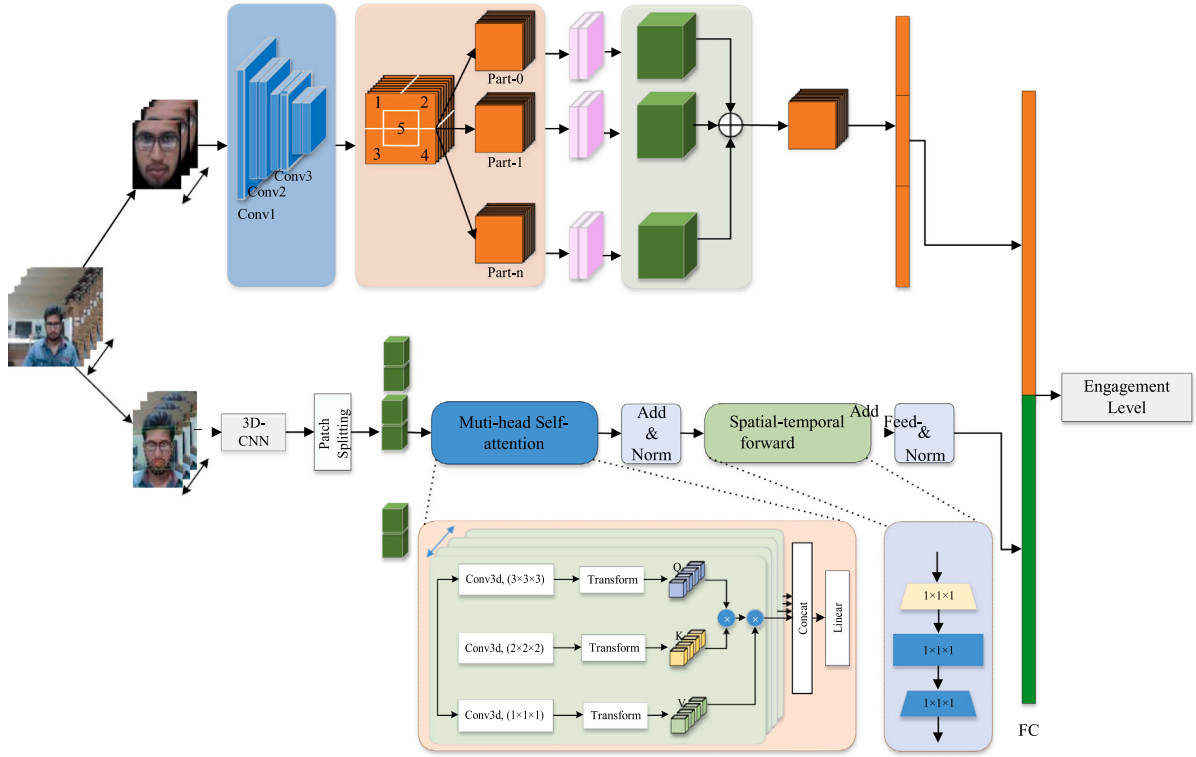


Fig. 1. Pipeline of the DTransformer.

This procedure enables the aggregation of spatio-temporal neighbor semantics and reduces computational costs for subsequent transformers. Specifically, with a targeted tube size of  $T_t \times H_t \times W_t$  (equivalent to the partition step size in non-overlapping settings), the resulting tube token map  $\mathbf{X}_{tube} \in \mathbb{R}^{D \times T \times H' \times W'}$  has dimensions of length, height, and width as defined by:

$$T' = \left\lfloor \frac{T}{T_s} \right\rfloor, H' = \left\lfloor \frac{H}{H_s} \right\rfloor, W' = \left\lfloor \frac{W}{W_s} \right\rfloor. \quad (4)$$

It is important to note that there is no inclusion of position embeddings after tube tokenization, as the 3D-CNN at the early stage already captures the relative spatio-temporal positions.

For the transformer, we elaborate the proposed architecture into two parts, i.e., multi-head attention and spatial-temporal feed-forward.

Within a transformer-based architecture as outlined in Ref. [38], the encoder module undertakes the processing of the input sequence to produce a comprehensive representation. For each token presented in the sequence, initial attention scores are computed by conducting a dot product operation among the query matrix  $Q$ , key matrix  $K$ , and value matrix  $V$ :

$$Att(Q, K, V) = Softmax\left(\frac{QK'}{\sqrt{d_k}}\right)V \quad (5)$$

In Eq. (9)  $d_k$  represents the dimensionality of the matrices  $Q$  and  $K$ . The resultant attention-weighted representation of the input sequence is attained through a weighted summation of the values  $V$  using the multi-head self-attention mechanism:

$$Multihead(Q, K, V) = Con(head_1, head_2, \dots, head_n)W \quad (6)$$

Here,  $n$  signifies the count of heads within the multi-head self-attention mechanism, and  $W$  denotes a weight matrix. This mechanism enables the model to concurrently focus on distinct segments of the input sequence while consolidating the resulting representations.

Following this, layer normalization (LN) is applied to the addition of the input  $\mathbf{X}_{tube}$  and the output from the multi-head attention mechanism, designated as  $MHD(Q, K, V)$ :

$$\mathbf{X}_{norm} = LN(\mathbf{X}_{tube} + MHD(Q, K, V)) \quad (7)$$

After the layer normalization operation, the subsequent step involves the utilization of a fully connected feed-forward network incorporating Rectified Linear Unit (ReLU) activation on the outcome obtained from the preceding stage, represented as  $textbf{X}_{norm}$ :

$$\mathbf{X}_{FFN} = ReLU(\mathbf{X}_{norm}W_1 + b_1)W_2 + b_2 \quad (8)$$

Thereafter, the output of the feed-forward network is added to  $\mathbf{X}_{tube}$ , and layer normalization is applied:

$$\mathbf{X}_{FFN} = LN(\mathbf{X}_{FFN} + \mathbf{X}_{tube}) \quad (9)$$

These operations are iterated  $N$  times, where  $N$  signifies the desired number of layers within the encoder.

## 4. Experiments

### 4.1. Datasets

The assessment of the algorithms was conducted upon two distinct data repositories known for capturing spontaneous engagement, specifically, the DAiSEE [6] and EmotiW-EP [39] collections. The former repository, DAiSEE, is composed of 9068 video segments, each with a duration of 10 s, featuring 112 participants. These segments were recorded in high definition — 1920 by 1080 pixels at a rate of 30 frames per second using a high-definition web camera. The recordings document the participants' level of attention to computer-based educational content within various unstructured settings. Expert annotations categorize these segments into quartet emotional states: engagement, boredom, confusion, and frustration, with each state further delineated into four intensity tiers: very low, low, high, and extremely high. The classification of intensity levels is inclusive of behavioral, cognitive, and emotional facets of engagement. In our research, we exclusively employ data corresponding to the 'engaged' state.

Subsequently, the EmotiW-EP dataset, a component of the Engagement Prediction segment of the EmotiW contest, encompasses samples from 78 subjects, 25 female and 53 male, ranging from 19 to 27 years in



**Table 1**

Performances of different architecture on the DAiSEE dataset.

| Methods                 | Accuracy (%) |
|-------------------------|--------------|
| A: PANet                | 63.8         |
| B: STformer             | 63.2         |
| A + B: PANet + STformer | 64.0         |

**Table 2**

Performances of different architecture on the EmotiW-EP dataset.

| Methods                 | MSE    |
|-------------------------|--------|
| A: PANet                | 0.1123 |
| B: STformer             | 0.0938 |
| A + B: PANet + STformer | 0.0729 |

age. The dataset comprises 262 video instances, distributed as follows: 147 for training, 48 for validation, and 67 for testing purposes. These were captured with a Microsoft Lifecam of F2.0 aperture at 640 by 480-pixel resolution at 30 frames per second in various unregulated scenarios, such as workplaces, hospitality settings, public areas, and more. The level of engagement for each video sample is quantified into four categories and graded on a scale from 0 to 1, with '0' denoting complete disengagement and '1' signifying full engagement. For the purposes of our investigation, only the training and validation video samples were employed to benchmark the proposed methodologies.

#### 4.2. Evaluation metrics

Researchers have employed a variety of evaluative indices for gauging engagement intensity. In their studies, Gupta and collaborators treated the challenge as one of categorical discernment and opted for top-1 accuracy as the assessment criterion to gauge the proficiency of algorithmic solutions within the DAiSEE dataset. In concordance with their approach, we adopted this same index to corroborate the efficacy of our techniques. Nonetheless, we discern certain limitations in the aforementioned evaluative protocols. Drawing inspiration from the methodology delineated in [39], we further construe the task of engagement prediction as one of quantitative estimation and hence utilize the mean square error (MSE) metric to quantify the performance of our approach on both the EmotiW-EP and DAiSEE datasets. In our task, Pytorch toolkit is adopted. And learning rate is set to 0.01 with the weight decay is 0.05. Batch size is set to 32.

#### 4.3. Results

In this part, ablation experiments are first conducted on the EmotiW-EP and DAiSEE datasets, and a comparison with SOTA methods is made for engagement detection.

##### 4.3.1. Ablation study

To further valid the efficacy of the proposed architecture, ablation experiments are conducted. Tables 1 and 2 show the performances for the proposed architecture. To make a clear comparison for the proposed architecture, we denote A as PANet, B as STformer, A+B as the proposed architecture. As in Table 1, one can note that A obtains the comparable performance for engagement detection with an accuracy of 63.8%, B obtains a bit performance with an accuracy of 63.2%. However, the fusion of A and B obtains the best performance for learner engagement detection with an accuracy of 64.0%. In Table 2, on the EmotiW-EP dataset, MSE is used to evaluate the accuracy for learner engagement detection. As Table 2 shows, one can note that, A achieves the accuracy of 0.1123, B obtains the accuracy of 0.0938, A + B obtains the best accuracy of 0.0729. Based on the analysis above, we can note that our introduced architecture obtains the competitive performances for learner engagement detection.

**Table 3**

Performances of different architecture on the DAiSEE dataset.

| Methods                              | Accuracy (%) |
|--------------------------------------|--------------|
| Video-level InceptionNet [6]         | 46.40        |
| Frame-level InceptionNet [6]         | 47.10        |
| C3D feature extraction [6]           | 48.6         |
| I3D [24]                             | 52.35        |
| ResNet + TCN with weighted loss [40] | 53.70        |
| C3D fine tuning [6]                  | 56.10        |
| C3D (focal loss) [41]                | 56.20        |
| LRCN [6]                             | 57.90        |
| DFSTN [42]                           | 58.84        |
| C3D + TCN [40]                       | 59.97        |
| DERN [25]                            | 60.00        |
| ResNet + LSTM [40]                   | 61.15        |
| ResNet + TCN [40]                    | 63.90        |
| 3D DenseAttNet [43]                  | 63.59        |
| Ours                                 | 64.00        |

**Table 4**

Performances of different architecture on the EmotiW-EP dataset.

| Methods            | MSE                        |
|--------------------|----------------------------|
| ResNet-LSTM [44]   | 0.0849 (val)               |
| ResNet-GRU [29]    | 0.1354 (val)               |
| C3D [45]           | 0.0934 (val)               |
| I3D [46]           | 0.1059 (val)               |
| EmotiW-EP baseline | 0.1000 (val) 0.1500 (test) |
| Chang et al. [47]  | 0.0441 (val) 0.0813 (test) |
| Thomas et al. [48] | 0.0655 (val) 0.0792 (test) |
| Niu et al. [28]    | 0.0671 (val) 0.0724 (test) |
| Yang et al. [30]   | 0.0398 (val) 0.0626 (test) |
| Wu et al. [49]     | 0.0350 (val) 0.0611 (test) |
| Zhu et al. [50]    | 0.0512 (val) 0.0541 (test) |
| Abedi et al. [51]  | 0.0673 (val)               |
| Kamath et al. [52] | 0.0450 (val) 0.0598 (test) |
| Liao et al. [53]   | 0.0736 (train+val)         |
| Mehta et al. [43]  | 0.0974 (train+val)         |
| Ours               | 0.0729 (val)               |

##### 4.3.2. Comparison with SOTA methods

To further explain the effectiveness of the introduced method, we compare it with the SOTA method in Tables 3 and 4. As Table 3 shows that the proposed architecture obtains the best performance for engagement detection.

The performance evaluation of the three proposed models, along with the comparative outcomes on the DAiSEE dataset, is presented in Table 3. Within [6], the authors delineated five distinct models: the video-level InceptionNet, frame-level InceptionNet, C3D feature extraction, C3D fine-tuning, and LRCN. The study reveals that LRCN achieves the highest accuracy of 57.9% among these models. Notably, LRCN surpasses other competitive models, including I3D [24], ResNet TCN with sampling and weighted loss [40], and C3D with focal loss [41]. The DFSTN [42] exhibits a performance of 58.84% on the DAiSEE dataset, surpassing LRCN by 0.94%. Additionally, DERN [25] achieves an accuracy of 60%, surpassing DFSTN by 1.16% and C3D + TCN [40] by 0.3%. The 3D DenseAttNet [43] achieves an accuracy of 63.59%. Furthermore, Abedi and Khan [40] introduces ResNet + TCN with sampling and weighted loss, C3D + TCN, ResNet + LSTM, and ResNet + TCN models, which respectively attains accuracies of 53.7%, 59.97%, 61.15%, and 63.9%. Notably, their study reports the highest accuracy of 61.15% and 63.9% among these models, surpassing DERN by 1.15% and 3.9%, respectively. Based on the above results, our proposed architecture obtains the best performance with the accuracy of 64.00% for learner engagement detection.

The performance evaluation of the three proposed models, along with the comparative outcomes on the EmotiW-EP dataset, is presented in Table 4. For these methods, we only adopt validation set for the ablation experiment. From the results, we can observe that the proposed method obtains the best performance with 0.0729. However, another method also obtains the better results for engagement detection.

Based on the results above, one can note that, the presented method obtains the excellent performance for engagement detection. Firstly, the 2D behaviors can be modeled by the 2D PANet. Secondly, the 3D features are modeled by the transformer based architecture. Thirdly, the combination of 2D and 3D features is modeled further for the engagement detection task.

## 5. Conclusion

In this paper, we propose a hybrid architecture, termed DTransformer, for engagement detection. Specifically, the architecture consists of two branches: the first is PANet, which can leverage the 2D-CNN to learn the discriminative features; the second is the transformer-based architecture, which can capture 3D features. Overall, the combination of PANet and STformer can learn the discriminative features of learners engagement level detection. Extensive experiments are conducted on DAiSEE and EmotiW-EP dataset, and obtain the competitive performances for learner engagement detection. In the future, we will explore the deep features for learner engagement detection.

## CRedit authorship contribution statement

**Rui Su:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Lang He:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Mengnan Luo:** Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## References

- [1] O. Mohamad Nezami, M. Dras, L. Hamey, D. Richards, S. Wan, C. Paris, Automatic recognition of student engagement using deep learning and facial expression, in: *Joint European Conference on Machine Learning and Knowledge Discovery in Databases*, Springer, 2020, pp. 273–289.
- [2] A. Dhall, G. Sharma, R. Goecke, T. Gedeon, EmotiW 2020: Driver gaze, group emotion, student engagement and physiological signal based challenges, in: *Proceedings of the 2020 International Conference on Multimodal Interaction*, 2020, pp. 784–789.
- [3] P. Guhan, M. Agarwal, N. Awasthi, G. Reeves, D. Manocha, A. Bera, ABC-net: Semi-supervised multimodal GAN-based engagement detection using an affective, behavioral and cognitive model, 2020, arXiv preprint arXiv:2011.08690.
- [4] A. Belle, R.H. Hargraves, K. Najarian, et al., An automated optimal engagement and attention detection system using electrocardiogram, *Comput. Math. Methods Med.* 2012 (2012).
- [5] K. Doherty, G. Doherty, Engagement in HCI: conception, theory and measurement, *ACM Comput. Surv.* 51 (5) (2018) 1–39.
- [6] A. Gupta, A. D'Cunha, K. Awasthi, V. Balasubramanian, Daisee: Towards user engagement recognition in the wild, 2016, arXiv, arXiv preprint arXiv:1609.01885.
- [7] X. Chen, L. Niu, A. Veeraraghavan, A. Sabharwal, FaceEngage: robust estimation of gameplay engagement from user-contributed (YouTube) videos, *IEEE Trans. Affect. Comput.* 13 (2) (2019) 651–665.
- [8] G. Zhao, M. Pietikainen, Dynamic texture recognition using local binary patterns with an application to facial expressions, *IEEE Trans. Pattern Anal. Mach. Intell.* 29 (6) (2007) 915–928.
- [9] J. Whitehill, Z. Serpell, Y.-C. Lin, A. Foster, J.R. Movellan, The faces of engagement: Automatic recognition of student engagement from facial expressions, *IEEE Trans. Affect. Comput.* 5 (1) (2014) 86–98.
- [10] S.K. D'Mello, S.D. Craig, J. Sullins, A.C. Graesser, Predicting affective states expressed through an emotive-aloud procedure from AutoTutor's mixed-initiative dialogue, *Int. J. Artif. Intell. Educ.* 16 (1) (2006) 3–28.
- [11] H.L. O'Brien, E.G. Toms, The development and evaluation of a survey to measure user engagement, *J. Am. Soc. Inf. Sci. Technol.* 61 (1) (2010) 50–69.
- [12] M. Cocea, S. Weibelzahl, Disengagement detection in online learning: Validation studies and perspectives, *IEEE Trans. Learn. Technol.* 4 (2) (2010) 114–124.
- [13] T. Aluja-Banet, M.-R. Sancho, I. Vukic, Measuring motivation from the virtual learning environment in secondary education, *J. Comput. Sci.* 36 (2019) 100629.
- [14] H. Monkarese, N. Bosch, R.A. Calvo, S.K. D'Mello, Automated detection of engagement using video-based estimation of facial expressions and heart rate, *IEEE Trans. Affect. Comput.* 8 (1) (2016) 15–28.
- [15] S.H. Fairclough, L. Venables, Prediction of subjective states from psychophysiology: A multivariate approach, *Biol. Psychol.* 71 (1) (2006) 100–110.
- [16] A.B. Khedher, I. Jraidt, C. Frasson, Tracking students' mental engagement using EEG signals during an interaction with a virtual learning environment, *J. Intell. Learn. Syst. Appl.* 11 (01) (2019) 1–14.
- [17] J.F. Grafsgaard, J.B. Wiggins, K.E. Boyer, E.N. Wiebe, J.C. Lester, Automatically recognizing facial indicators of frustration: a learning-centric analysis, in: *2013 Humaine Association Conference on Affective Computing and Intelligent Interaction*, IEEE, 2013, pp. 159–165.
- [18] G. Littlewort, J. Whitehill, T. Wu, I. Fasel, M. Frank, J. Movellan, M. Bartlett, The computer expression recognition toolbox (CERT), in: *2011 IEEE International Conference on Automatic Face & Gesture Recognition*, FG, IEEE, 2011, pp. 298–305.
- [19] N. Bosch, S. D'Mello, R. Baker, J. Ocumpaugh, V. Shute, M. Ventura, L. Wang, W. Zhao, Automatic detection of learning-centered affective states in the wild, in: *Proceedings of the 20th International Conference on Intelligent User Interfaces*, 2015, pp. 379–388.
- [20] M. Saneiro, O.C. Santos, S. Salmeron-Majadas, J.G. Boticaio, et al., Towards emotion detection in educational scenarios from facial expressions and body movements through multimodal approaches, *Sci. World J.* 2014 (2014).
- [21] A. Kamath, A. Biswas, V. Balasubramanian, A crowdsourced approach to student engagement recognition in e-learning environments, in: *2016 IEEE Winter Conference on Applications of Computer Vision*, WACV, IEEE, 2016, pp. 1–9.
- [22] N. Dalal, B. Triggs, Histograms of oriented gradients for human detection, in: *2005 IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, CVPR'05, Vol. 1, IEEE, 2005, pp. 886–893.
- [23] A. Schulz, J.F. Cohn, J. Shen, M. Pantic, Automatic measurement of visual attention to video content using deep learning, in: *2019 16th International Conference on Machine Vision Applications*, MVA, IEEE, 2019, pp. 1–6.
- [24] H. Zhang, X. Xiao, T. Huang, S. Liu, Y. Xia, J. Li, An novel end-to-end network for automatic student engagement recognition, in: *2019 IEEE 9th International Conference on Electronics Information and Emergency Communication*, ICEIEC, IEEE, 2019, pp. 342–345.
- [25] T. Huang, Y. Mei, H. Zhang, S. Liu, H. Yang, Fine-grained engagement recognition in online learning environment, in: *2019 IEEE 9th International Conference on Electronics Information and Emergency Communication*, ICEIEC, IEEE, 2019, pp. 338–341.
- [26] A. Dhall, A. Kaur, R. Goecke, T. Gedeon, EmotiW 2018: Audio-video, student engagement and group-level affect prediction, in: *Proceedings of the 20th ACM International Conference on Multimodal Interaction*, 2018, pp. 653–656.
- [27] A. Dhall, EmotiW 2019: Automatic emotion, engagement and cohesion prediction tasks, in: *2019 International Conference on Multimodal Interaction*, 2019, pp. 546–550.
- [28] X. Niu, H. Han, J. Zeng, X. Sun, S. Shan, Y. Huang, S. Yang, X. Chen, Automatic engagement prediction with gap feature, in: *Proceedings of the 20th ACM International Conference on Multimodal Interaction*, 2018, pp. 599–603.
- [29] J. Chung, C. Gulcehre, K. Cho, Y. Bengio, Empirical evaluation of gated recurrent neural networks on sequence modeling, 2014, arXiv preprint arXiv:1412.3555.
- [30] J. Yang, K. Wang, X. Peng, Y. Qiao, Deep recurrent multi-instance learning with spatio-temporal features for engagement intensity prediction, in: *Proceedings of the 20th ACM International Conference on Multimodal Interaction*, 2018, pp. 594–598.
- [31] C. Thomas, S. Purvaj, D.B. Jayagopi, Student engagement from video using unsupervised domain adaptation, in: *IMPROVE*, 2022, pp. 118–125.
- [32] T. Selim, I. Elkabani, M.A. Abdou, Students engagement level detection in online e-learning using hybrid efficientnetb7 together with tcn, lstm, and bi-lstm, *IEEE Access* 10 (2022) 99573–99583.
- [33] S.M. Villaroya, J.J. Gamboa-Montero, A. Bernardino, M. Maroto-Gómez, J.C. Castillo, M.Á. Salichs, Real-time engagement detection from facial features, in: *2022 IEEE International Conference on Development and Learning*, ICDL, IEEE, 2022, pp. 231–237.
- [34] A. Karpathy, G. Toderici, S. Shetty, T. Leung, R. Sukthankar, L. Fei-Fei, Large-scale video classification with convolutional neural networks, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2014, pp. 1725–1732.
- [35] J. Lee, W. Reade, R. Sukthankar, G. Toderici, et al., The 2nd youtube-8m large-scale video understanding challenge, in: *Proceedings of the European Conference on Computer Vision (ECCV) Workshops*, 2018.

- [36] M. Monfort, A. Andonian, B. Zhou, K. Ramakrishnan, S.A. Bargal, T. Yan, L. Brown, Q. Fan, D. Gutfreund, C. Vondrick, et al., Moments in time dataset: one million videos for event understanding, *IEEE Trans. Pattern Anal. Mach. Intell.* 42 (2) (2019) 502–508.
- [37] T. Xiao, M. Singh, E. Mintun, T. Darrell, P. Dollár, R. Girshick, Early convolutions help transformers see better, *Adv. Neural Inf. Process. Syst.* 34 (2021) 30392–30400.
- [38] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A.N. Gomez, Ł. Kaiser, I. Polosukhin, Attention is all you need, *Adv. Neural Inf. Process. Syst.* 30 (2017).
- [39] A. Kaur, A. Mustafa, L. Mehta, A. Dhall, Prediction and localization of student engagement in the wild, in: 2018 Digital Image Computing: Techniques and Applications, DICTA, IEEE, 2018, pp. 1–8.
- [40] A. Abedi, S.S. Khan, Improving state-of-the-art in detecting student engagement with resnet and tcn hybrid network, in: 2021 18th Conference on Robots and Vision, CRV, IEEE, 2021, pp. 151–157.
- [41] L. Geng, M. Xu, Z. Wei, X. Zhou, Learning deep spatiotemporal feature for engagement recognition of online courses, in: 2019 IEEE Symposium Series on Computational Intelligence, SSCI, IEEE, 2019, pp. 442–447.
- [42] J. Liao, Y. Liang, J. Pan, Deep facial spatiotemporal network for engagement prediction in online learning, *Appl. Intell.* 51 (2021) 6609–6621.
- [43] N.K. Mehta, S.S. Prasad, S. Saurav, R. Saini, S. Singh, Three-dimensional DenseNet self-attention neural network for automatic detection of student's engagement, *Appl. Intell.* 52 (12) (2022) 13803–13823.
- [44] L.S.-T. Memory, Long short-term memory, *Neural Comput.* 9 (8) (2010) 1735–1780.
- [45] S. Ji, W. Xu, M. Yang, K. Yu, 3D convolutional neural networks for human action recognition, *IEEE Trans. Pattern Anal. Mach. Intell.* 35 (1) (2012) 221–231.
- [46] J. Carreira, A. Zisserman, Quo vadis, action recognition? a new model and the kinetics dataset, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017, pp. 6299–6308.
- [47] C. Chang, C. Zhang, L. Chen, Y. Liu, An ensemble model using face and body tracking for engagement detection, in: Proceedings of the 20th ACM International Conference on Multimodal Interaction, 2018, pp. 616–622.
- [48] C. Thomas, N. Nair, D.B. Jayagopi, Predicting engagement intensity in the wild using temporal convolutional network, in: Proceedings of the 20th ACM International Conference on Multimodal Interaction, 2018, pp. 604–610.
- [49] J. Wu, B. Yang, Y. Wang, G. Hattori, Advanced multi-instance learning method with multi-features engineering and conservative optimization for engagement intensity prediction, in: Proceedings of the 2020 International Conference on Multimodal Interaction, 2020, pp. 777–783.
- [50] B. Zhu, X. Lan, X. Guo, K.E. Barner, C. Boncelet, Multi-rate attention based gru model for engagement prediction, in: Proceedings of the 2020 International Conference on Multimodal Interaction, 2020, pp. 841–848.
- [51] A. Abedi, S. Khan, Affect-driven engagement measurement from videos, *Computer* 11 (2021) 12.
- [52] S. Kamath, P. Singhal, G. Jeevan, B. Annappa, Engagement analysis of students in online learning environments, in: Machine Learning and Big Data Analytics (Proceedings of International Conference on Machine Learning and Big Data Analytics (ICMLBDA) 2021), Springer, 2022, pp. 34–47.
- [53] E. Sanchez, M.K. Tellamekala, M. Valstar, G. Tzimiropoulos, Affective processes: stochastic modelling of temporal context for emotion and facial expression recognition, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2021, pp. 9074–9084.